TimeTensors Experiment 3: Training Losses

1 Theoretical overview and goal

Raw MSE weights large-scale series and outliers strongly; MAE is more robust. For look-back scale $s(x) = \mathrm{sd}(x) + \varepsilon$, normalized losses use

$$\mathrm{nMSE} = H^{-1} \sum_h \left(\frac{\hat{y}_h - y_h}{s(x)} \right)^2, \qquad \mathrm{nMAE} = H^{-1} \sum_h \frac{|\hat{y}_h - y_h|}{s(x)}.$$

Relative MSE instead scales by $|\,\mathrm{mean}(x)| + \varepsilon$. These objectives change which users and windows dominate learning even when all final models are evaluated in the same data-space MSE.

Experimental goal. Determine whether scale-balanced or robust optimization improves generalization, especially across heterogeneous users, without changing the reported evaluation metric.

2 Implementation details

The sweep compares MSE, MAE, nMSE, nMAE, and relative MSE. Loss selection is a training-only change; output denormalization and test MSE are identical across methods. Small denominators are clamped by the implementation epsilon.

The default protocol uses six datasets, three settings, DLinear, seeds 1–3, random sampling, batch size 256, learning rate 10^{-5}, 10,000 optimizer steps, and 1,000-step validation/logging. PatchTST is run after the sampling, constant-user, and normalization decisions have been fixed. The full profile retains the larger two-model sweep.

Launcher: `05_losses.slurm`

Tables should report data-space MSE, equal-user MSE, and W10 when available, with mean $\pm$ seed standard deviation and row scaling. Training curves must be compared in their native objective units only; nMSE and MSE curve heights are not directly commensurate.

3 Interpretation

If normalized training improves equal-user or W10 metrics more than ordinary MSE, the gain comes from reweighting heterogeneous series. A raw-MSE gain with worse W10 suggests concentration on high-scale or easy users.